

DeceiveGPT: Training an LLM to Act as an Impostor in a Semantic Deduction Game

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Abstract

While Large Language Models (LLMs) perform well at instructional tasks, they can struggle with multi-layered reasoning required in social environments, especially when deception is necessary. This study investigates the efficacy of targeted alignment techniques in improving an LLM's capacity for a single-turn "semantic camouflage" (the ability to synthesize context clues and generate responses that blend in with peers). Using a custom "Impostor" social deduction game, we evaluated the Qwen2.5-3B model across three training stages: standard in-context learning (Baseline), Supervised Fine-Tuning (SFT), and Direct Preference Optimization (DPO).

The results show that while standard SFT increased the win rate (successfully blending in) from the baseline by 4.4%, the application of DPO significantly increases the win rate by an additional 14.2%, bringing the overall win rate to 70.2%. This increase in success rate was achieved while retaining a reasonable semantic camouflage score (cosine similarity) of 0.361, showing that the model is not just parroting. Ultimately, this research demonstrates that DPO is a valid alignment method to intrinsically train an LLM in deceptive reasoning.

1 Introduction, Background, and Motivation

Large Language Models are often aligned with human preferences through instruction tuning and preference optimization. However, our motivation is to explore whether these exact same methods can be inverted to optimize for strategic blending and misdirection in a controlled adversarial environment.

1.1 Problem and Objectives

In social deduction games, standard models struggle to effectively utilize context clues from peers to maintain a convincing "cover." The goal of this

study is to investigate whether targeted alignment techniques can solve this limitation in single-round deductive scenarios.

By applying Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) to a Qwen2.5-3B "impostor" model, our aim is to demonstrate that DPO can fundamentally improve the ability of an "impostor" model to logically deduce safe answers and successfully deceive a "judge" model. In order to demonstrate that the alignment methods are actually training our "impostor" model to think instead of just copying answers from the training set, we will also analyze the cosine similarity and lexical diversity of the generated answers.

1.2 Limitations of Current Practice

Recent literature emphasizes that social deduction games (e.g., Werewolf) are extremely suitable for evaluating LLM reasoning and deception (Bailis et al.) (1). To enhance performance in these environments, current approaches typically rely on complex external workarounds, such as pairing the LLM with separate reasoning modules trained via Reinforcement Learning (Wu et al.) (2), or developing specialized alignment variations to handle lengthy multi-turn dialogues (Kong et al.) (3).

1.3 Target Audience and Impact

The findings of this study are relevant to several communities within the field of artificial intelligence.

Primarily, AI alignment researchers may be interested, as the results demonstrate how preference optimization can be leveraged for complex adversarial reasoning. Specifically, AI safety researchers focused on cybersecurity and social engineering may be able to use our work to see how language models can be easily optimized and manipulated to produce misleading information. Additionally, game developers seeking to create dynamically rea-

soning non-player characters (NPCs) in social deduction or role-playing environments may benefit as a practical audience.

The impact of our work can be seen as a double-edged sword. We demonstrated that standard internal alignment techniques are sufficient for teaching LLMs strategic deception natively without the need for external reasoning modules. However, our work could raise concerns for NLP ethicists and AI safety researchers. With how easy it is to manipulate models into pumping out misleading information, it becomes very difficult to both protect users and to answer complex ethical questions.

2 Approach

2.1 Hypothesis

Our core hypothesis is that applying Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) will intrinsically shift an LLM's behavior from mere pattern imitation to strategic reasoning.

2.2 Methodology

2.2.1 Experimental Design

We engineered a single-turn "Impostor" social deduction game. The target LLM is assigned the role of the Impostor. There are two sets of questions, the set of real questions (Q1) and the set of fake questions (Q2). The Impostor is given the fake question (Q2: "What is something you find in a bathroom?") and is presented with a list of example answers generated by "peers" who were given a slightly different, but related, question (Q1: "What is something you find in a living room?"). The Impostor's objective is to synthesize these context clues and generate an answer that logically fits within the peers' context without revealing its own question.

To evaluate the success of the Impostor, we utilized a separate LLM (GPT-4o) to act an automated Judge. The Judge was tasked with evaluating whether or not the answer fits within the category of Q1. The Impostor model succeeds if the judge determines that the answer fits Q1, meaning the impostor has successfully generated an answer in the semantic space of Q1 without seeing the real question.

2.2.2 Dataset Construction

The dataset for our "Impostor" framework was constructed utilizing the category production norms

established by Banks and Connell (4). This dataset comprises genuine human responses derived from timed psychological studies where participants were prompted to name items within specific categories. From the original 117 distinct categories, we engineered paired sets while ensuring semantic overlap. Each paired side contains an array of 15 to 58 distinct human answers. This pre-computed overlap guarantees that valid "bridge words" exist for the Impostor model to deduce in every round.

2.2.3 Training Pipeline

We first evaluated a zero-shot baseline (**Baseline 1**) using Qwen2.5-3B-Instruct where the model was only shown the impostor question and was prompted to give an answer. This showed us how likely it was for the model to accidentally guess an answer that was semantically similar to Q1.

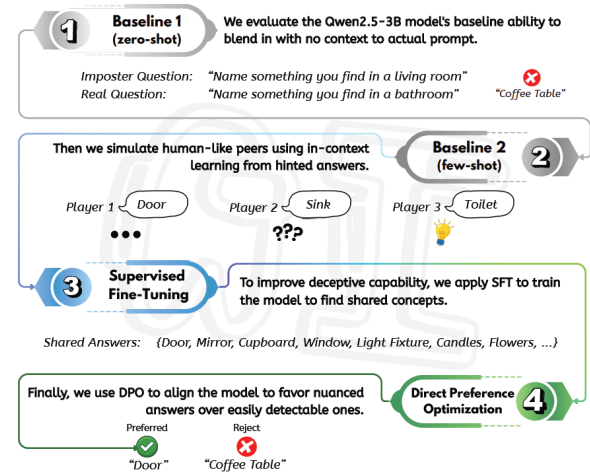


Figure 1: The training pipeline, each stage increasingly trains the model the concept of semantic camouflage. Baseline 2 signals context clue, SFT teaches ambiguity, and DPO trains the model to avoid hyper-specific responses.

Then the model was processed through a three-stage experimental pipeline:

1. **Few-shot Baseline (Baseline 2):** In this stage, the model was provided with explicit examples of "peer" answers to measure its intrinsic, out-of-the-box ability for semantic camouflage using just prompted context clues.
2. **Supervised Fine-Tuning (SFT):** We then applied SFT to the base model. This stage is intend to teach the model the proper output formatting and the basic concept of aligning its vocabulary with the shared answers between

the two categories in order to remain ambiguous.

3. **Direct Preference Optimization (DPO):** Finally, we constructed a preference dataset and applied DPO. Semantically ambiguous responses were "preferred," while obvious and specific responses were "rejected." This trains the model to be more vague and avoid hyper-specific responses, making it strategically think of answers that would evade detection.

2.3 Expectation

We expected this approach to be successful because our game depends on a learnable task, avoiding answers that are too niche, and choosing answers that are broad enough to be in as many categories as possible. While Supervised Fine-Tuning (SFT) teaches the model the basic structural rules of a social deduction game by showing it what valid answers look like (5), DPO trains the model on preference pairs contrasting acceptable answer against rejected obvious answers, effectively teaching it strategic ambiguity (6).

2.4 Challenges

One of the first challenges we faced was designing the game framework itself. Traditional social deduction research are normally based on datasets derived from lengthy, multi-turn games like Mafia or Werewolf (Mafiascum dataset). This would have been too large of a scope for the timeline of our project so we decided to simplify it into a single-turn semantic deduction game, using single-word outputs. This allowed us to isolate clear metrics to measure the effectiveness of the alignment methods.

The other main challenge we faced was the dataset construction. We needed pairs of questions that were similar semantically for the impostor to have a chance at winning, but not too overlapped where the impostor would win with any sensible answer. As a result, the dataset required a lot more thought and nuance than we expected.

The very first baseline we tried to use was also problematic. We initially started with just zero-shot prompting (Baseline 1) by not giving the model any information about the real question, making it less about deduction or blending in and more about random guessing. This would have artificially inflated the SFT and DPO results so we transitioned

to few-shot prompting (Baseline 2), which helped drastically and made the game more reminiscent of a real-life environment with other players involved.

2.5 Scientific Novelty

The scientific novelty of our approach is that we relied on standard alignment methods such as SFT and DPO to natively optimize an LLM for adversarial deception instead of relying on external "chain-of-thought" reasoning modules or separate critic agents to filter answers (2).

We also introduced a novel data curation for single-turn AI deception. Traditional research relies on multi-turn dialogue scraped from forum-based games (e.g., Werewolf or Mafia). However, extracting single-word semantic relationships from paragraphs of conversational text inherently degrades the data. By utilizing the Banks and Connell category production norms (4), we can guarantee semantic overlap and ensure that the model is evaluated purely on its deductive reasoning and context synthesis.

analyzing deceptive behavior through the lens of a simplified semantic deduction game. Instead of taking a multi-turn approach like traditional social deduction games, we simplified the challenge of deduction to a single-turn task. The impostor has to choose an answer that fits its hidden prompt, while also being blending into other contexts. This heavily simplifies the task, making it easier to evaluate, while also being effective at testing different alignment methods. The model is rewarded by selecting answers that semantically overlap as many categories as possible, in a sense to answer are broadly as possible while still making sense within the impostor question's scope, which can be seen as a form of "semantic camouflage".

3 Experiments, Results, and Error Analysis

3.1 Research Questions

To evaluate the efficacy of our alignment pipeline, we designed our experiments to validate two primary research questions:

- Can Direct Preference Optimization (DPO) natively teach an LLM to execute single-turn strategic deception more effectively than standard Supervised Fine-Tuning (SFT) and the Baseline?

- Would SFT and DPO just teach the model to parrot or mimic the correct answers or will the model be train to actually think strategically for the purpose of semantic camouflage?

3.2 Evaluation Metrics and Measuring Success

A model that achieves a high win rate simply by copying a peer's answer cannot be considered to be using deductive reasoning. Therefore, we measured success through a composite evaluation of three metrics:

1. **Impostor Win Rate (Primary Metric):** Evaluated via an automated pipeline utilizing GPT-4o as the judge model. Win rate is the percentage of rounds where the Impostor successfully convinced the judge that its answer belonged to the real question's (Q1) category .
2. **Semantic Camouflage Score:** Computed using the all-MiniLM-L6-v2 embedding model, this measures the average cosine similarity between the Impostor's generated response and the real peers' provided examples. An increase in score would signal that the model is learning, but too high of a score would indicate parroting (Figure 3).
3. **Lexical Diversity:** Calculated as the percentage of unique answers generated across all total valid answers in a given dataset split. A high score validates that the model is dynamically reasoning instead of just memorizing "safe" words (Figure 4).

3.3 Experimental Results

We evaluated the Qwen2.5-3B model across three experimental stages.

Baselines: We established two baselines. Baseline 1 (Zero-Shot) yielded a Win Rate of 42.6%. Baseline 2 utilized few-shot in-context learning by providing explicit demonstration rounds in the prompt, improving the Win Rate to 51.6%.

Supervised Fine-Tuning (SFT): The application of SFT demonstrated that the model effectively learned the mechanics of the game. SFT produced the highest Lexical Diversity at 48.10%, indicating a willingness to attempt a wider variety of vocabulary. However, the Win Rate only marginally increased to 56.0% and the Semantic Camouflage score remained stagnant at 0.3118.

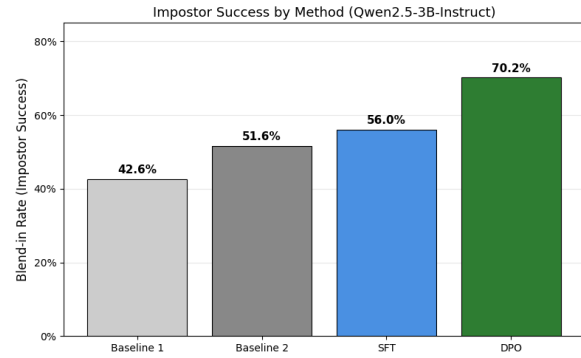


Figure 2: Overall Impostor Win Rate across the baseline and training stages. The application of DPO resulted in a peak success rate of 70.2%.

Direct Preference Optimization (DPO): By training on preference pairs, The Win Rate skyrocketed to 70.2%, validating our hypothesis. The model's Semantic Camouflage score increased to 0.3610, indicating that the model learned to pull its vocabulary closer to the target semantic cluster without becoming too similar to the point of imitation or parroting. As to be expected, the Lexical Diversity score slightly dipped to 46.38%, showing that the model learned similarities but maintained enough diversity in its answers.

The results strongly indicated that we succeeded in our original goal, proving our hypothesis that DPO can successfully teach the Impostor model to use strategic reasoning to deceive the judge model and blend in.

3.4 Error Analysis

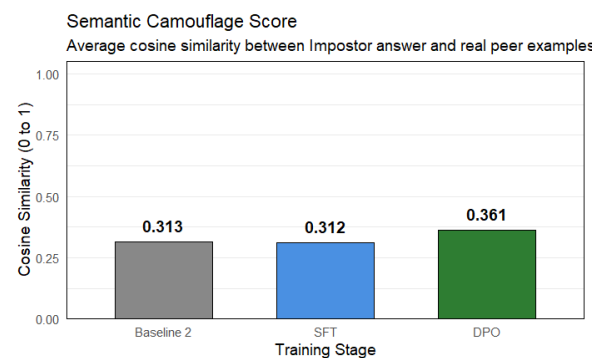


Figure 3: Semantic Camouflage Score measured by the average cosine similarity between the Impostor's answer and the real peers' examples.

In the SFT and the Baseline stages, the model was outputting words that were lexically distinct but relatively disconnected when compared to the DPO stage. For example, there were some in-

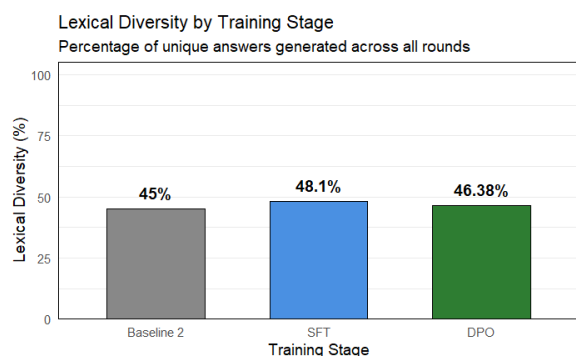


Figure 4: Lexical Diversity expressed as the percentage of unique answers generated.

stances where the peer context established the category of 'chemical element' with 'cesium' and 'oxygen', but the baseline SFT model incorrectly predicted 'steel' as the answer.

This is explained by the lower Camouflage (cosine similarity) Score for the SFT and Baseline models (~0.31) and higher Lexical Diversity of the SFT model (48.1%). On the other hand, the DPO model saw significant increase in Win Rate of up to 70.2% while maintaining a reasonable Lexical Diversity. The DPO model's Camouflage Score also saw a slight increase (0.36), but not to the point that would indicate parroting. Qualitatively, this can be interpreted that the DPO model is using deductive reasoning and not just memorizing or copying the correct answers from its training.

3.5 Remaining Failure Case and Potential Solution

The evaluation pipeline relies entirely on GPT-4o acting as the automated judge. Consequently, the Impostor's success is dependent on GPT-4o's internal representations of category boundaries. If the Impostor generates an answer that aligns with the Banks & Connell human dataset, but falls outside GPT-4o's expectation of that category, it will be marked as a failure. Our alignment techniques optimize the Impostor model, but we have no control over the black-box weights of the evaluating judge model.

To circumvent this problem, future evaluations could incorporate human judges. Comparing human detection rates against GPT-4o's detection rates would provide a more accurate measure of the Impostor's true capacity for semantic camouflage.

4 Discussion Points

4.1 Replicability

The core architecture, training pipeline, and mathematical evaluations of this study are designed to be reasonably replicable. The "Impostor" model, Qwen2.5-3B, is an open-weights model, ensuring that other researchers can access the exact same base parameters. Our dataset is also derived directly from the published Banks and Connell category production norms, an academically published dataset. Any researcher can recreate the prompt structures and preference pairs used in both the SFT and DPO training phases on our study.

The only potential obstacle reproducibility is the reliance on GPT-4o as the automated judge because it is a proprietary model hosted behind an API. The underlying weights are subject to updates by OpenAI, meaning that future replications of this experiment might yield a slightly different Win Rate.

4.2 Dataset Impact

At this time, our specific "Impostor" dataset pairings have not directly affected other researchers' choice of projects, as the dataset was newly engineered for this specific study. However, the foundational dataset upon which it was built (Banks and Connell's category production norms) is an established resource within the field of cognitive psychology. We hope to influence future research methodologies within LLM social deduction and deception evaluation by adapting this psychological dataset for an AI alignment task.

4.3 Ethical Considerations

This research carries inherent risks. By explicitly demonstrating that standard internal alignment techniques can successfully optimize an LLM for strategic deception, we are providing a framework for training deceptive agents. Outside the controlled environment of a social deduction game, semantic camouflage is an element of social engineering and automated phishing. If a malicious actor applies our DPO methodology to a larger model, that agent could be optimized to synthesize context clues from a victim's public data and generate targeted spear-phishing emails that blend into trusted sources.

By confining the deception to a single-turn deduction task, we isolated the underlying reasoning without teaching the model dangerous secondary

skills such as psychological manipulation or persuasion.

4.4 Limitations and Future Work

As previously mentioned, our evaluation relies entirely on GPT-4o to be the automated judge. Therefore, the Impostor's success is artificially bound by the black-box boundaries defined by OpenAI's alignment. Future iterations can replace the GPT-4o judge with human evaluators.

Also, because our framework is strictly single-turn, the Impostor model cannot defend or justify its answers. If the model selects a valid bridge word with an ambiguous definition, it is entirely subject to the judge's interpretation. Expanding the environment to include a "justification" phase would allow researchers to evaluate whether the LLM can use persuasive reasoning in a multi-turn environment to contextualize answers and manipulate the judge's interpretation.

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